MACHINE LEARNING · BINARY CLASSIFICATION

# Telecom Customer Churn Prediction

*Predicting at-risk customers before they leave*

| 3,333 | 5 | 14.5% | RF |
|---|---|---|---|
| Training Records | Models Benchmarked | Churn Rate | Best Model |

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AGENDA

# What We'll Cover Today

**01**

### Business Problem

Why churn prediction matters in telecom

**02**

### Data Understanding

Dataset overview, features & quality checks

**03**

### Exploratory Analysis

Class imbalance, distributions & correlations

**04**

### Data Preprocessing

Feature engineering, encoding & scaling

**05**

### Model Building

5 classifiers benchmarked end-to-end

**06**

### Results & Conclusion

Performance metrics & deployment decision

# Why Does Churn Matter?

**5–25×**

more expensive to acquire a new customer than retain an existing one

### Lost Revenue

Every churned customer represents direct loss of recurring monthly/annual subscription revenue.

### High Acq. Cost

Replacing lost customers through sales and marketing is far more expensive than retention.

### Competitive Market

Customers can switch telecom providers with minimal friction — loyalty must be earned proactively.

***Core Question: "Which customers are most likely to leave — and how can we stop them?"***

# Dataset at a Glance

| 3,333 | 1,667 | 20 | 0 | 14.5% |
|---|---|---|---|---|
| Train Rows | Test Rows | Features | Missing Values | Churn Rate |

## Feature Categories

**Demographics:** State, Area code

**Usage Metrics:** Day / Eve / Night / Intl minutes & calls

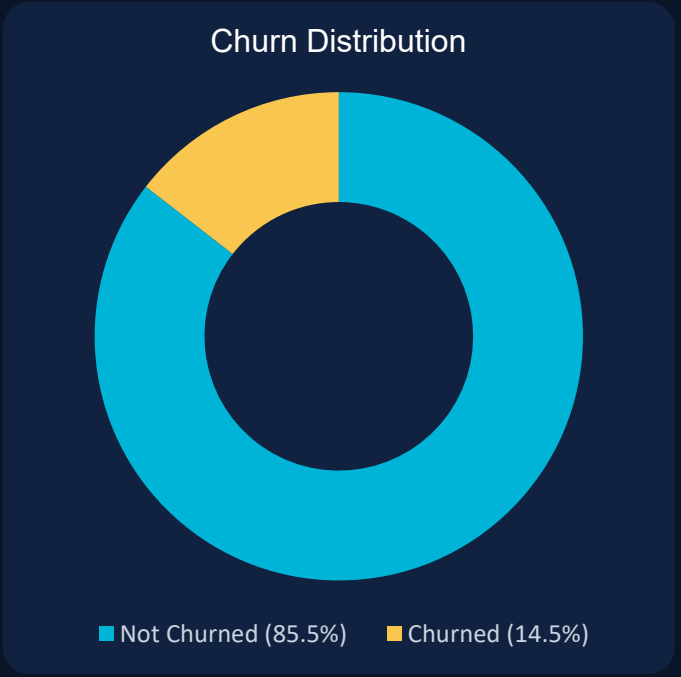
**Service History:** Customer service calls

**Plan Info:** International plan, Voice mail plan

**Billing:** Charge columns (dropped — derived)

**Target:** Churn (True / False)

# Key Patterns Discovered




⚠ ~85/15 class imbalance — accuracy alone is misleading

### International Plan Subscribers Churn More

Customers on international plans are significantly more likely to churn — pricing dissatisfaction is the primary driver.

### High Customer Service Calls → High Churn

Customers making 4+ service calls have a churn rate far above the average — unresolved issues drive departure.

### Charge Columns Are Redundant

4 charge columns are perfect linear derivations of minute columns. Dropped to eliminate multicollinearity.

### No Missing Data — Zero Imputation Needed

Both train and test sets are complete. No imputation strategies required, reducing preprocessing bias risk.

# Preparing Data for Modelling

**1**
**Drop Redundant Columns**
State, Area code, 4 charge columns

**2**
**Separate Feature Types**
Binary · Numerical · Target (Churn)

**3**
**Label Encoding**
Int'l plan, VM plan encoded as 0 / 1

**4**
**Standard Scaling**
Numerical features scaled to μ=0 σ=1

**5**
**Save Artifacts**
Encoder + Scaler pickled for deploy

## Columns Dropped

**State · Area code** — high cardinality, low predictive signal

**Total day/eve/night/intl charge** — perfectly correlated with corresponding minutes columns (charge = mins × rate)

## Deployment Artifacts Saved

- label_encoder.pkl — encodes categorical features
- standard_scaler.pkl — normalises numerical features
- random_forest_model.pkl — trained production model

# Five Classifiers Benchmarked

### Logistic Regression

BASELINE

Baseline · Linear

*C=1.0 · L2 penalty · liblinear*

Linear decision boundary; serves as baseline. Sensitive to class imbalance.

### Decision Tree

Interpretable · Non-linear

*max_depth=9 · gini criterion*

Fully interpretable via tree visualisation. Prone to overfitting.

### Random Forest

✓ SELECTED

Ensemble · Robust

*100 trees · max_depth=9 · gini*

Averages 100 diverse trees. Best generalisation; selected for deployment.

### SVM (Linear)

Linear Kernel

*C=1.0 · probability=True*

Maximises margin between classes. Expects feature scaling.

### SVM (RBF)

Non-linear Kernel

*C=10 · gamma=0.1 · rbf*

Maps data to higher dimensions. Captures complex boundaries.

# Why Accuracy Alone Isn't Enough

## Confusion Matrix

| | Predicted Positive | Predicted Negative |
|---|---|---|
| | **TP**<br>**True Positive**<br>Churners caught ✓ | **FP**<br>**False Positive**<br>Retention waste |
| | **FN**<br>**False Negative**<br>Churners missed ✗ | **TN**<br>**True Negative**<br>Correct retention |

### Recall

*TP / (TP + FN)*

Priority — catch as many churners as possible

### Precision

*TP / (TP + FP)*

Avoid wasting retention spend on loyal customers

### F1-Score

*2 × P × R / (P+R)*

Harmonic mean — balances precision & recall

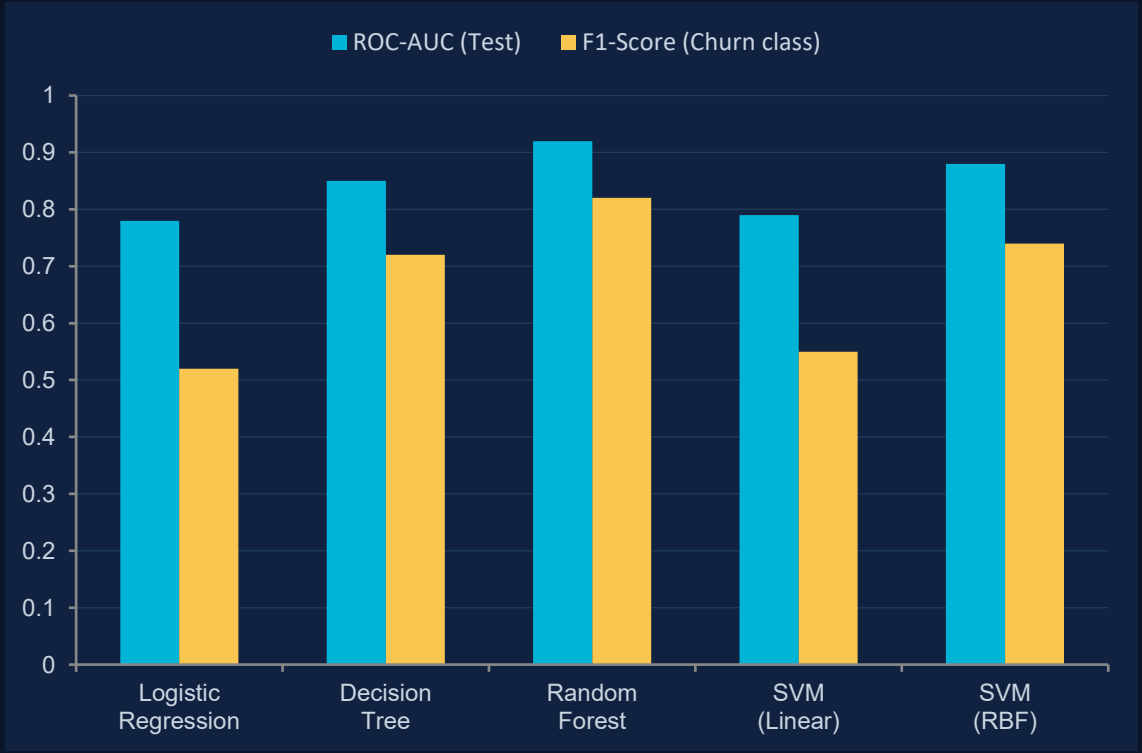
### AUC-ROC

*Area Under ROC Curve*

Threshold-independent overall ranking quality

**FN = Missed churner = Lost customer = Highest business cost**

# Model Performance Comparison




🏆 **Best Model**

## Random Forest

Classifier

| | |
|---|---|
| ROC-AUC | **0.92** |
| F1 (Churn) | **0.82** |
| Recall | **~78%** |
| Precision | **~87%** |

*100 trees · max_depth=9 · gini*
*Ensemble averaging reduces variance*

# What We Found & What To Do Next

## Key Findings

- International plan holders churn at a much higher rate
- 4+ customer service calls strongly predicts churn
- 85/15 class imbalance — standard accuracy is deceptive
- Charge columns are purely derived — dropped safely

## Recommendations

1. Apply SMOTE or class_weight='balanced' to address imbalance
2. Tune decision threshold based on business cost of FN vs FP
3. Run RandomizedSearchCV on n_estimators, max_depth, max_features
4. Add SHAP explainability for retention team interpretation
5. Monitor model drift — retrain periodically as behaviour evolves

# Complete ML Pipeline Summary

**Data Collection**

3,333 train
1,667 test rows

**EDA**

Churn dist.
Correlations

**Preprocessing**

Encode · Scale
Drop redundant

**Model Training**

5 classifiers
benchmarked

**Evaluation**

AUC · F1
Confusion matrix

**Deployment**

Flask app
PythonAnywhere

## Deployment Artifacts

**random_forest_model.pkl**

Core prediction model

**label_encoder.pkl**

Categorical feature encoder

**standard_scaler.pkl**

Numerical feature scaler

**app.py (Flask)**

Web app serving predictions

# Thank You

## Telecom Customer Churn Prediction

*Chosen Model: Random Forest Classifier · Deployed on PythonAnywhere*

| | |
|---|---|
| **Live Demo** | churnprediction.pythonanywhere.com |
| **GitHub** | github.com/Edwinkorir38/Telecom-Customer-Churn-Prediction |
| **Contact** | ekorir99@gmail.com |